INCH-POUND

MIL-PRF-1/466C
17 October 2008
SUPERSEDING
MIL-E-1/466B
4 February 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5842

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design after 7 March 1997.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Triode, miniature, medium Mu, grounded grid

Outline	---	6-1 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential
Base connections:		

Pin no.	---	1	2	3	4	5	6	7	8	9
Element	---	a	nc	h	g	g	k	g	g	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	RK	Ik	Pp	Ehk	Ck	TE	Alt
Unit:	V	V dc	V dc	Ohms	mA dc	W	V	µF	°C	ft
Maximum:	6.9	200	---	---	38	4.5	55	---	130	Note 2
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	150	0	60	---	---	---	Note 1	---	---

GENERAL:

Qualification – Not Required.

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TABLE I. Testing and inspection

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level 6/	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 1							
1256	Electrode current (1) (anode)	---		0.65	Ib	16.0	30.0	mA dc
1256	Electrode current (2) (anode)	---	Ec = -10 V dc	0.65	Ib	---	100	µA dc
1266	Total grid current	3,4		0.65	Ic	0	-0.5	µA dc
1306	Transconductance (1)	---		0.65	Sm	20,000	30,000	µmhos
1306	Transconductance (2)	4	Ef = 5.7 V	0.65	ΔSm	---	12	%
1201	Short and discontinuity detection	---		0.4	---	---	---	---
	Conformance inspection, part 2							
1211	Insulation of electrodes	---		4.0	---	---	---	---
1031	Low-frequency vibration	---	Rp = 2,000 ohms; Rk = 0; Ec = -2.0 V dc	---	Ep	---	500	mV ac
1336	Heater-cathode leakage	---		---	Ihk	---	20	µA dc
1301	Heater current	---		---	If	280	320	mA
1316	Amplification factor	---		---	Mu	35	55	---
1331	Direct-interelectrode capacitance	---	Without external shield	6.5	Ck-gh Ck-kh Cp-gh	8.0 --- 1.5	10.0 0.55 1.95	pF pF pF
2126	Glass strain	---		4.0	---	---	---	---
1121	Base strain	---		---	---	---	---	---
1105	Permanence of marking	---		---	---	---	---	---
	Conformance inspection, part 3							
1501	Intermittent life	5	Group A; Ehk = 55 V; TE =130°C	---	---	---	---	---
---	Intermittent life-test end points (500 hours):							
1306	Transconductance (1)	---		---	Sm	16,000	---	µmhos
1266	Total grid current	---		---	Ic	---	-1.0	µA dc

NOTES:

1. The reactance of the cathode by-pass capacitor shall be 0.5-ohm maximum at the test frequency.
2. See “Reduced pressure (altitude) rating” paragraph in the basic document.
3. Approved alternate test: The insertion of a 0.47 megohm resistor in series with the grid shall not change the anode current by more than 3.0 mA dc.
4. This test to be performed at the conclusion of the holding period.
5. Envelope temperature (ET) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
6. This specification sheet uses accept on zero sampling in accordance with MIL-PRF-1, table III.

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Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project No. 5960-2008-063)

Review activities:
Army - AR, CR4
Navy - AS, CG, MC, OS
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

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